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In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = sns.load_dataset('iris')
df.columns = ['c1','c2','c3','c4','c5']
df.head()
```

Out[1]:

	c1	c2	c3	c4	c5
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

In [3]:
```
df.tail()
```

Out[3]:

	c1	c2	c3	c4	c5
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

In [4]:
```
df.describe()
```

Out[4]:

	c1	c2	c3	c4
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   c1      150 non-null    float64
 1   c2      150 non-null    float64
 2   c3      150 non-null    float64
 3   c4      150 non-null    float64
 4   c5      150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [6]:
```
df.isnull().sum()
```

Out[6]:
```
c1    0
c2    0
c3    0
c4    0
c5    0
dtype: int64
```

In [7]:
```
df.shape
```

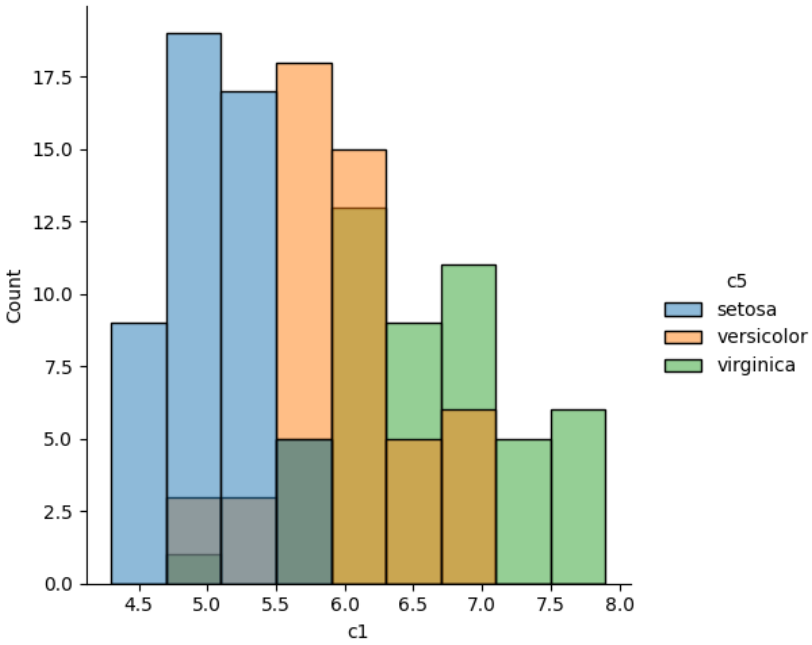
Out[7]:
```
(150, 5)
```

In [8]:
```
df.columns
```

Out[8]:
```
Index(['c1', 'c2', 'c3', 'c4', 'c5'], dtype='object')
```

In [9]:
```
sns.displot(x = 'c1' , data = df , hue = 'c5')
```

Out[9]:
```
<seaborn.axisgrid.FacetGrid at 0x7720b812b2e0>
```



In [13]:
```
import matplotlib.pyplot as plt

df.hist(bins = 20 , figsize = (10,8))
plt.suptitle('Historgram of Iris Fetaures')
plt.show()

#case 2. Histogram (Feature Distribution)
#Histogram for Each Feature
#Explanation
#A histogram shows how values are distributed.  Helps understand: Range
```

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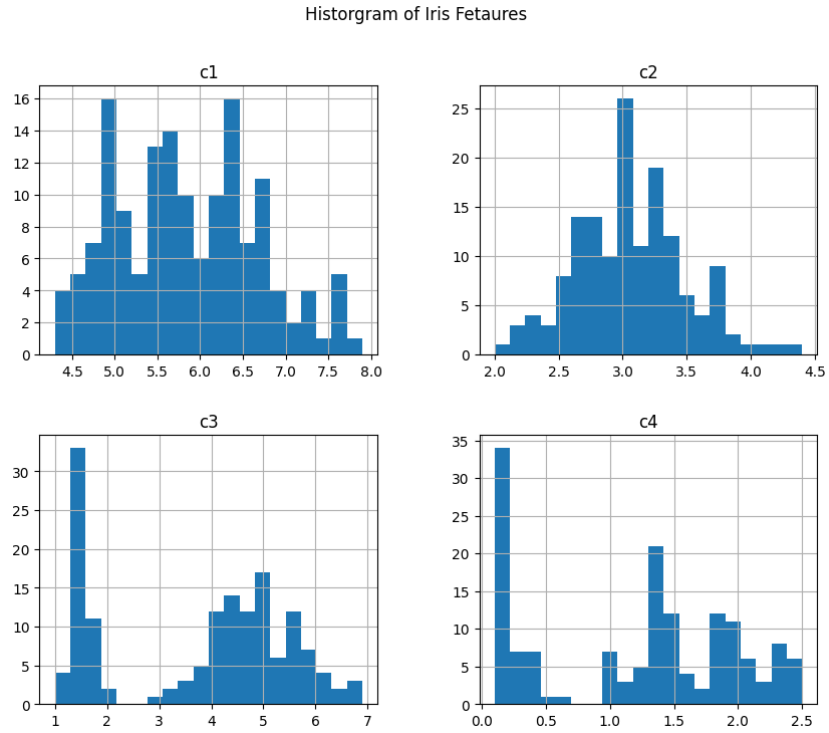
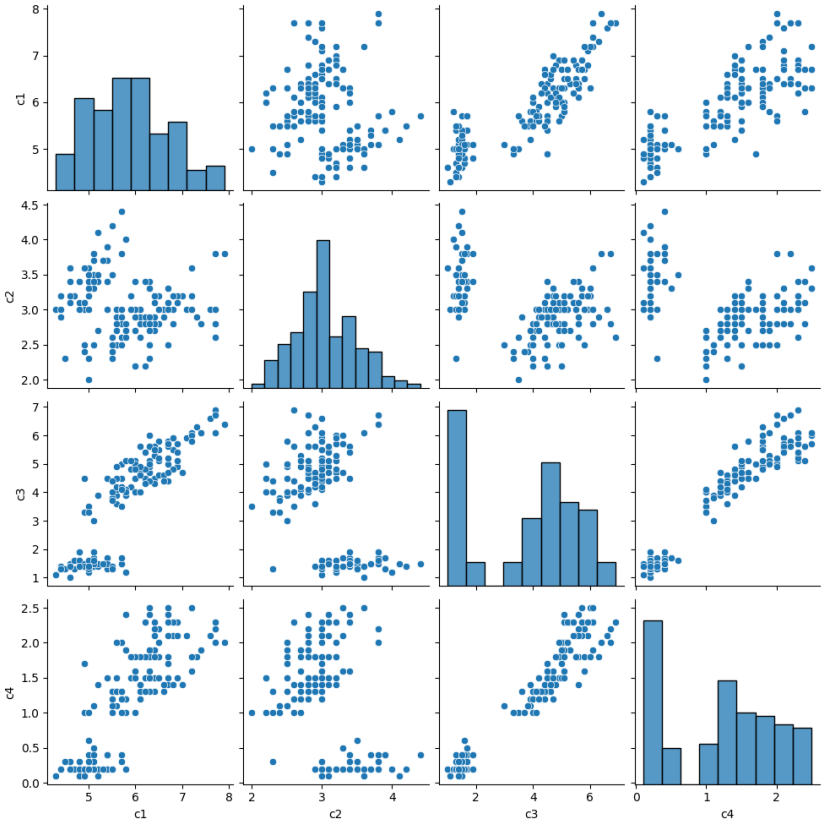
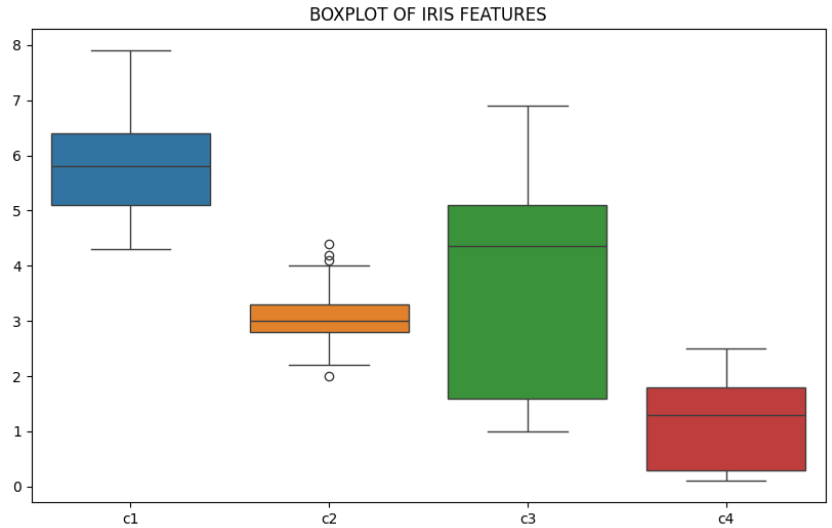


In [14]:
```
sns.pairplot(df)
plt.show()
```





In [17]:
```
plt.figure(figsize = (10,6))

sns.boxplot(data = df)
plt.title("BOXPLOT OF IRIS FEATURES")
plt.show()
#case 3- Interpretation -Points outside whiskers = Outliers
#Petal features usually have fewer outliers
#Sepal width may show some extreme values
```

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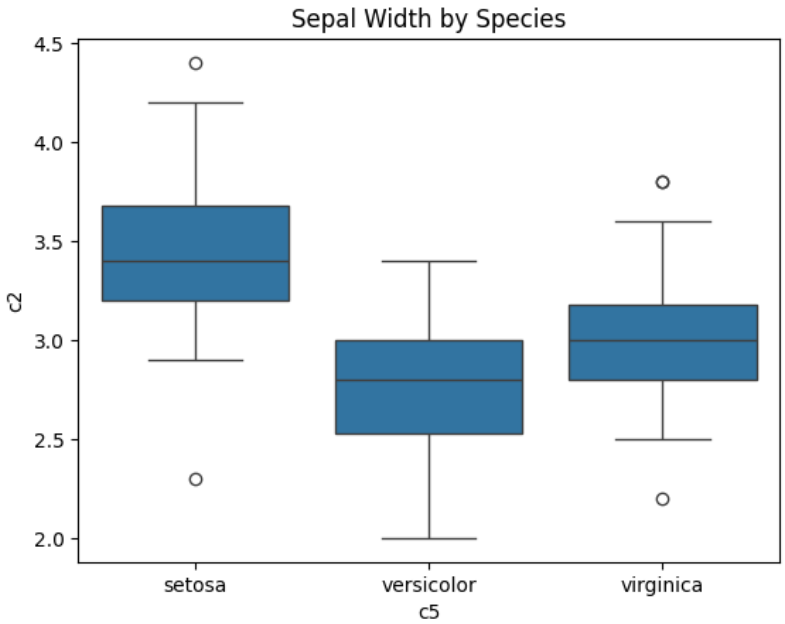
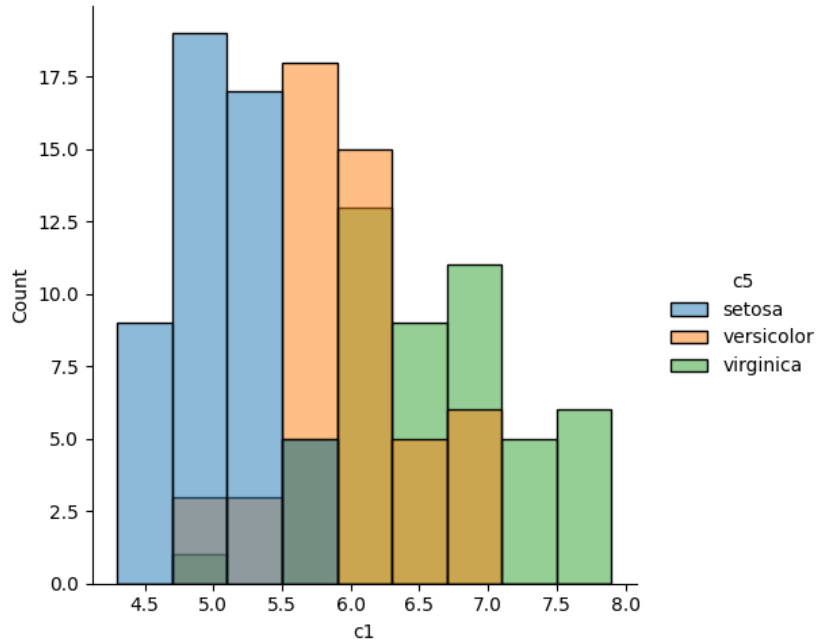
In [18]:
```
# Boxplot grouped by species
sns.boxplot(x ='c5', y ='c2', data = df)
plt.title("Sepal Width by Species")
plt.show()

#Petal Length & Width Clearly separate species → very important features
#Sepal Length-Moderate overlap
#Sepal Width-Most variation + some outliers
#Outliers-Visible in sepal width,--Very few in petal features
```



In [19]:
```
sns.displot(x='c1',data=df,hue='c5')
```

Out[19]: <seaborn.axisgrid.FacetGrid at 0x77204ae27130>



In [20]:
```
sns.displot(x ='c1',data = df)

#Inference:

 #   In the histogram for sepal length, we observe that the distribution
 #   Most of the iris flowers have a sepal length ranging from approximat
  #  There are relatively fewer flowers with extremely short or long sepa
```

Out[20]: <seaborn.axisgrid.FacetGrid at 0x772049bfa110>

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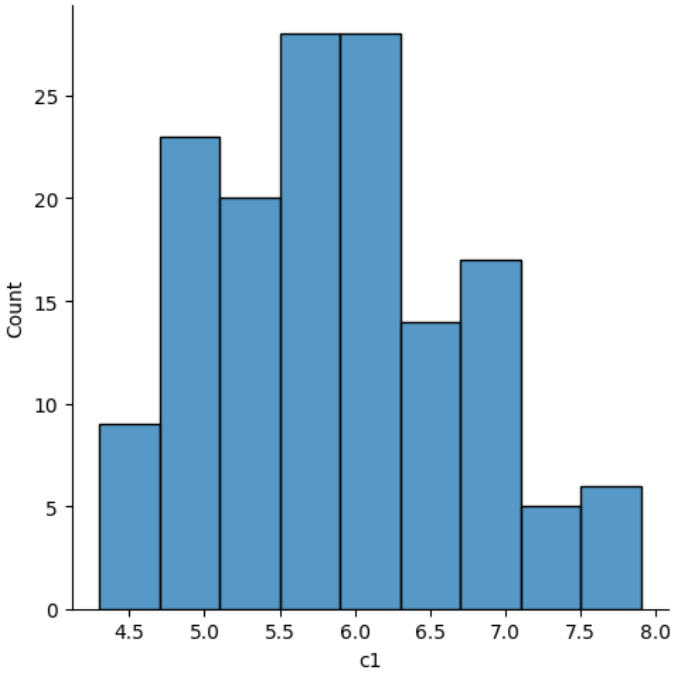
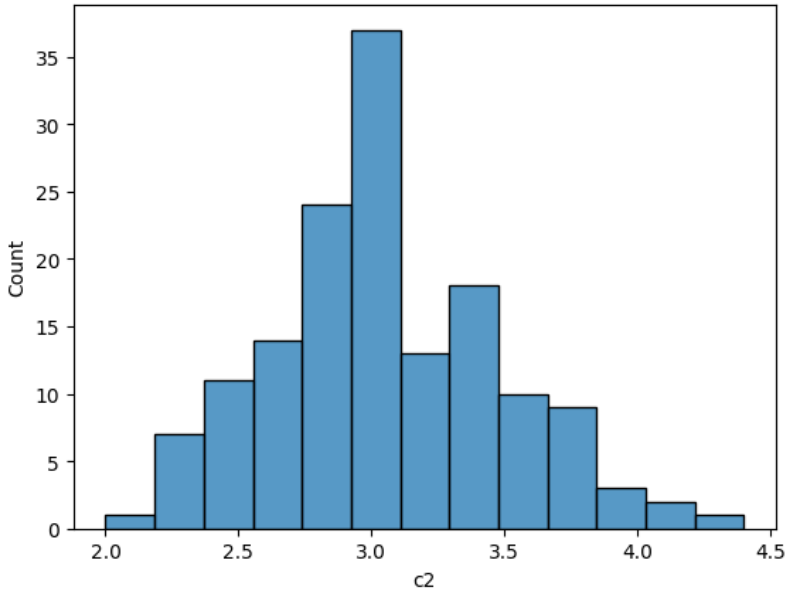


In [21]:
```
sns.histplot(x ='c2',data = df)

#Inference:
#    The histogram for sepal width shows a roughly normal distribution, a
 #   The most common sepal width appears to be around 3.0 cm.
  #  There is a noticeable spread in sepal widths, ranging from approxima
  #there's a slight skew towards higher sepal widths, as indicated by the
  #Inference: Shows th distribution of the sepal_width of the iris flower
```

Out[21]: <Axes: xlabel='c2', ylabel='Count'>



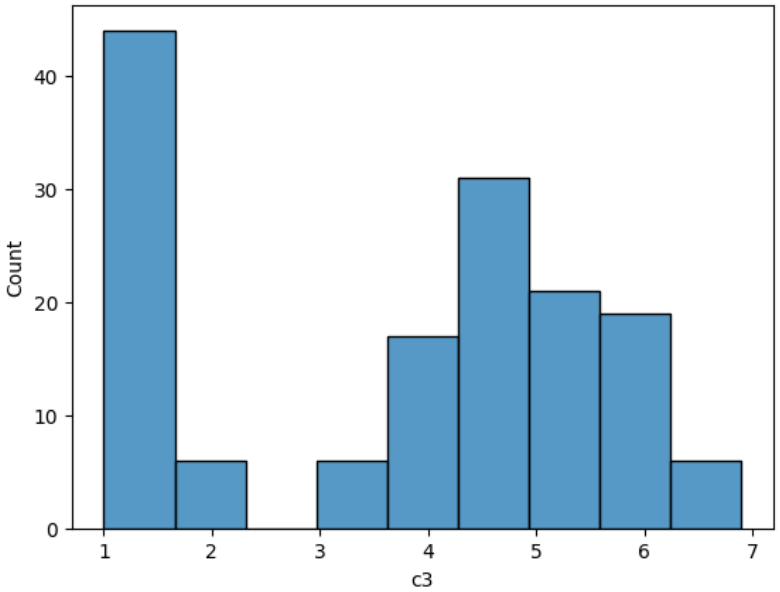
In [22]:
```
sns.histplot(x ='c3',data = df)

#Inference:

 #   Data is not normally distributed
  #  Some of the petal length are ranging from 1-2
   # Most data is in range of 3-7


   #Data is not normally distributed
#Some of the petal length are ranging from 0-0.5
#Most data is in range of 1.0-2.5
```

Out[22]: <Axes: xlabel='c3', ylabel='Count'>



In [ ]: